

FIELD DYE-PENETRANT EXAMINATION LOG

Unit: Hydrocarbon Reflux Plant | Technician ID: T-409 | Date: 2026-08-16

OFFLINE FIELD NDT
VERIFIED LOG

EXAMINATION METHOD: Color Contrast Solvent Removable Liquid Penetrant (ASTM E165)

TARGET COMPONENT: Reflux Pump 3B Lower Volute Weld Seam W-202

SURFACE PREPARATION: Wire brushed and degreased with solvent cleaner.

FIELD OBSERVATIONS & DEFECT INDICATION:

- Linear defect indication detected along longitudinal weld seam W-202.
- Surface breaking fatigue crack confirmed extending 48 mm along seam axis.
- Crack aperture measures approximately 1.4 mm at widest opening.
- Minor weld seam porosity and cluster pitting observed in surrounding heat-affected zone (HAZ).

TECHNICIAN VERDICT:

CRITICAL SEAM FRACTURE: Crack depth extends beyond allowable cosmetic grind limit.

Recommends visual photo confirmation and immediate replacement of casing.

CERTIFIED INSPECTOR SIGN-OFF

Sign: J. Miller (Level II PT/UT)

Status: REJECT ■ REPLACE REQUIRED